

Pneumonia-Oriented Disease Prediction Using Multinomial Logistic Regression

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Abstract

This paper presents a disease-focused extension of the supplied healthcare machine-learning project, selecting pneumonia as the example disease. The source project provides a general multi-class disease diagnosis framework based on symptoms and patient-profile information, including fever, cough, fatigue, difficulty breathing, age, gender, blood pressure, and cholesterol level. Multinomial Logistic Regression is retained as the primary classifier. Since the supplied files do not contain a pneumonia-labelled raw dataset or numerical experimental results, no unsupported accuracy or diagnostic-performance values are reported.

Keywords: Pneumonia; Multinomial Logistic Regression; Disease Prediction; Healthcare; Machine Learning; Clinical Decision Support

1. Introduction

Pneumonia is selected as a disease-focused application of the healthcare prediction framework. The supplied healthcare presentation describes a general multi-class disease diagnosis system based on symptoms and patient-profile information rather than a pneumonia-only dataset. This paper therefore specializes the framework to pneumonia as the selected disease case.

The source project lists fever, cough, fatigue, and difficulty breathing among the symptoms, together with age, gender, blood pressure, and cholesterol level as patient-profile information.

1.1 Objectives

- Develop a pneumonia-oriented disease prediction framework.
- Use the symptom and patient-profile features documented in the project.
- Apply Multinomial Logistic Regression.
- Provide a structured workflow for preprocessing, prediction, and evaluation.

2. Background and Motivation

The supplied research references demonstrate the value of statistical analysis for complex datasets. The first paper systematically reviews PCA-based variable reduction, while the second applies PCA to high-dimensional player-tracking data. For the healthcare application, the project instead selects Multinomial Logistic Regression because the target task is disease classification.

3. Disease-Focused Feature Design

Feature group	Variables	Role
Symptoms	Fever, cough, fatigue, difficulty breathing	Primary symptom indicators
Demographics	Age, gender	Patient context
Clinical profile	Blood pressure, cholesterol level	Additional profile information
Target	Disease category	Classification outcome

Table 1. Feature design for the pneumonia-oriented case study.

4. Exploratory Data Analysis

The EDA follows the project workflow: checking dataset structure, identifying missing values and duplicates, analyzing disease/class distribution, studying age and gender distribution, and examining relationships between input features.

4.1 Data Quality

Before model fitting, incomplete records and duplicate entries should be identified. Categorical variables such as gender should be encoded numerically.

4.2 Class Distribution

Disease-class distribution should be examined before training because class imbalance can influence the interpretation of classification metrics. The supplied files do not provide actual class counts.

5. Method

5.1 Preprocessing

Input records are cleaned, categorical variables are converted into numerical representations, and selected features are organized into a model-ready matrix. The data are then divided into training and testing subsets.

5.2 Multinomial Logistic Regression

The classifier estimates the probability of each disease class from the patient feature vector. The category with the highest estimated probability is selected as the preliminary prediction.

5.3 Workflow

Step	Description
Patient data	Collect symptoms and profile information
Cleaning	Handle missing values and duplicates
Encoding	Convert categorical variables
Feature preparation	Select and organize model inputs
Train-test split	Create training and testing subsets
Model fitting	Train Multinomial Logistic Regression
Prediction	Estimate the disease category
Evaluation	Accuracy, precision, recall, F1-score

Table 2. Pneumonia-oriented prediction workflow.

6. Proposed Pneumonia-Oriented Application

In this focused case, pneumonia is treated as the selected disease within the broader multi-class framework. A patient record containing fever, cough, fatigue, difficulty breathing and profile variables is processed and supplied to the trained classifier. The output is a preliminary disease-category estimate.

Because the source project does not provide a pneumonia-labelled raw dataset, this paper does not claim that the supplied data have been experimentally validated for pneumonia prediction.

7. Results and Discussion

The project specifies accuracy, precision, recall, and F1-score for evaluation but provides no numerical results. Consequently, this paper reports the methodological outcome rather than inventing performance values.

8. Limitations

- The supplied project does not identify pneumonia as an existing target class.
- The raw healthcare dataset is not supplied.
- No numerical accuracy, precision, recall, or F1-score is reported.
- Clinical validation is not provided.

9. Conclusion

A pneumonia-oriented disease case study was formulated from the general healthcare machine-learning framework supplied in the project. The study uses the documented symptoms and patient-profile variables and retains Multinomial Logistic Regression as the classification method.

10. Future Scope

- Obtain a disease-specific clinically labelled dataset.
- Add clinically meaningful respiratory and laboratory features.

- Compare Multinomial Logistic Regression with other machine-learning methods.
- Use cross-validation and class-balance strategies.
- Develop a web or mobile prototype with appropriate clinical safeguards.

11. Author Contributions and Declarations

Conceptualization, disease-focused adaptation, methodology, and writing: M. Geethika Chowdary and G. Varshini. Academic guidance: Shaik Asif. Funding, conflicts of interest, and clinical validation were not specified in the supplied files.

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